

Assignment – 22:

Problem Statement:

Write a menu driven program in C to perform various string operations using in-built functions.

Code:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str1[100], str2[100];
    int choice;
    printf("Enter first string: ");
    scanf("%s", str1);
    printf("Enter second string: ");
    scanf("%s", str2);
    printf("\n--- MENU ---\n");
    printf("1. Length of string\n");
    printf("2. Copy string\n");
    printf("3. Compare strings\n");
    printf("4. Concatenate strings\n");
    printf("5. Reverse string\n");
    printf("6. Substring check\n");
    printf("7. Exit\n");
    do
    {
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
```

case 1:

```
printf("Length of first string = %lu\n", strlen(str1));  
break;
```

case 2:

```
strcpy(str2, str1);  
printf("After copying, second string = %s\n", str2);  
break;
```

case 3:

```
if (strcmp(str1, str2) == 0)  
{  
    printf("Strings are equal\n");  
}  
else  
{  
    printf("Strings are not equal\n");  
}  
break;
```

case 4:

```
strcat(str1, str2);  
printf("Concatenated string = %s\n", str1);  
break;
```

case 5:

```
{  
    char temp[100];  
    strcpy(temp, str1);  
    int len = strlen(temp);  
    for (int i = 0; i < len / 2; i++)  
    {  
        char ch = temp[i];
```

```

        temp[i] = temp[len - i - 1];
        temp[len - i - 1] = ch;
    }
    printf("Reversed string = %s\n", temp);
    break;
}

case 6:
    if (strstr(str1, str2) != NULL)
    {
        printf("Substring found\n");
    }
    else
    {
        printf("Substring not found\n");
    }
    break;

case 7:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice\n");
}
} while (choice != 7);
}

```

Input & Output:

```
Enter first string: Hello
Enter second string: World
```

```
--- MENU ---
```

1. Length of string
2. Copy string
3. Compare strings
4. Concatenate strings
5. Reverse string
6. Substring check
7. Exit

```
Enter your choice: 1
Length of first string = 5
Enter your choice: 3
Strings are not equal
Enter your choice: 6
Substring not found
Enter your choice: 7
Exiting program...
```

```
Enter first string: Wonderful
Enter second string: World
```

```
--- MENU ---
```

1. Length of string
2. Copy string
3. Compare strings
4. Concatenate strings
5. Reverse string
6. Substring check
7. Exit

```
Enter your choice: 4
Concatenated string = WonderfulWorld
Enter your choice: 5
Reversed string = dlrowlufrednoW
Enter your choice: 7
Exiting program...
```

```
Enter first string: Wonder
Enter second string: Woman
```

```
--- MENU ---
```

1. Length of string
2. Copy string
3. Compare strings
4. Concatenate strings
5. Reverse string
6. Substring check
7. Exit

```
Enter your choice: 2
After copying, second string = Wonder
Enter your choice: 4
Concatenated string = WonderWonder
Enter your choice: 7
Exiting program...
```

